



Republic of the Philippines
CAVITE STATE UNIVERSITY
College of Engineering and Information Technology
Don Severino delas Alas Campus
Indang, Cavite

SOFTWARE ENGINEERING I PROJECT TECHNICAL DOCUMENTATION

Date Submitted

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Project Manager

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Members

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Contact Person

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The Company

Company Name:	Redeemed Route Logistics Inc.
Company Address:	3rd Floor, Glass Door, AGDC Commercial Bldg, Trece Martires - Indang Rd, Inocencio, Trece Martires City, 4109 Cavite
Major Products/Services:	- Trucking Services – Provides trucks for transporting goods and products from one location to another based on client booking requests, ensuring safe and scheduled delivery of cargo. - Cargo Delivery Services – Handles the delivery of client goods, materials, or products to designated destinations through company-owned trucks. - Truck Booking Services – Allows clients to reserve and schedule available trucks for specific delivery dates, routes, and transportation needs. - Freight Transportation Services – Offers transportation solutions for bulk or commercial goods requiring large-capacity vehicles for business deliveries.

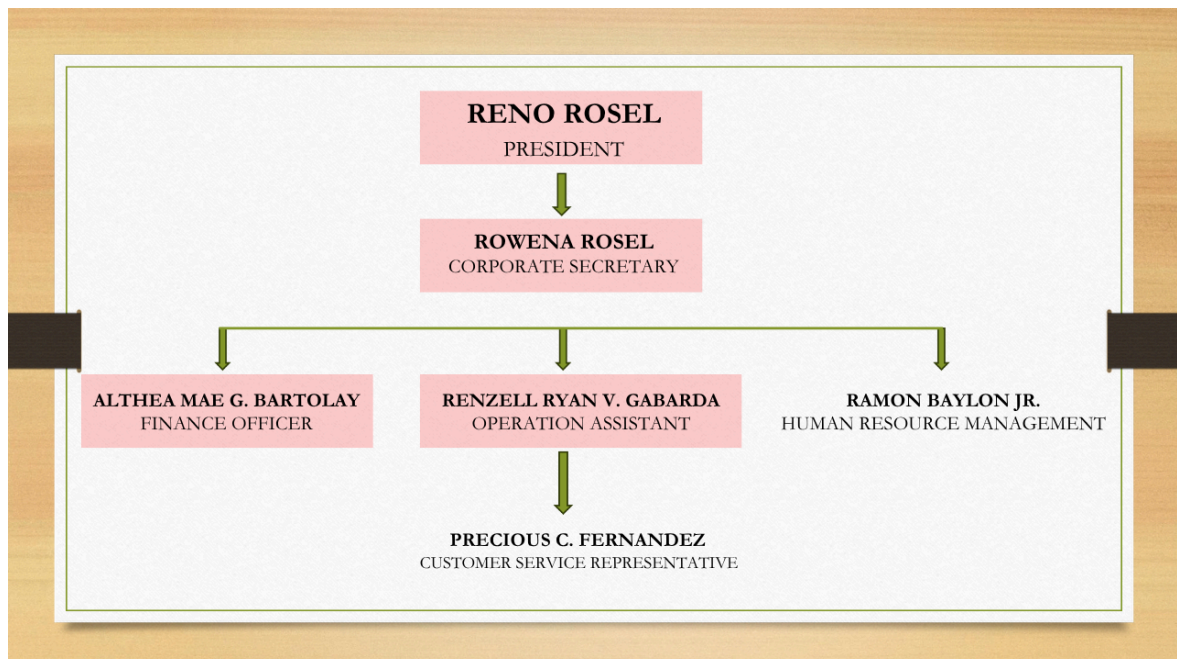
- Minor Products/Services:
- **Delivery Scheduling Services** – Assists clients in arranging preferred delivery dates and times for more organized transportation operations.
 - **Route Coordination Services** – Plans and manages delivery routes to ensure goods are transported efficiently to their intended destinations.
 - **Dispatching Services** – Assigns available trucks and coordinates delivery operations based on client booking requirements.

Nature of the Proposed System

- Draw the interrelationship of the proposed system with the other systems in the company. Highlight/Color the proposed system.

At present, **the company does not utilize any formal or integrated information systems in its operations, finance, or other business functions.** Based on the follow-up interview, most records and processes are managed manually using Microsoft Excel and basic bookkeeping practices rather than dedicated software systems. Since there are no existing operational or accounting systems currently implemented within the company, there is no established system interrelationship that can be illustrated in relation to the proposed system.

- Department or section in the organization/company where the proposed system is being used or will be used. Likewise draw the organizational chart of the company, and indicate by shading/coloring the department which uses the proposed system.



III. Implementation of the proposed system
a. Partially automated

Draw the different modules and sub-modules in the proposed system that is automated/computerized. Use color codes when necessary. Include a discussion of each module and how it is being used.



The proposed RouteX system begins when an authorized user records a delivery transaction by entering the necessary delivery information into the system. These details include the delivery ID, truck plate number, assigned driver, client information, delivery route, delivery date, delivery status, revenue, and delivery-related expenses. The system automatically computes the delivery margin based on the recorded revenue and expenses.

All recorded information is stored in the system database and becomes available for monitoring, retrieval, updating, and reporting purposes. Users can search records using plate numbers, delivery IDs, drivers, or client names through the system's global search functionality. The system also allows authorized users to update delivery records, upload Proof of Delivery (POD) documents, and maintain accurate operational records.

In addition to delivery transactions, the system allows authorized personnel to record company expenses that are not directly associated with a specific delivery. Examples include insurance payments, vehicle registration fees, office supplies, utilities, equipment purchases, and other business expenses. These records are stored separately to support more accurate profitability analysis and financial reporting.

The system provides reporting functionalities that generate operational and financial reports such as delivery activity reports, revenue reports, expense reports, and profitability reports. These reports support management decision-making and provide visibility into the company's operational and financial performance.

Access to the system is protected through authentication and role-based access controls. Authorized users are assigned specific permissions based on their responsibilities, while user activities are recorded in audit logs to maintain accountability and security.

1. Delivery Management Module

The Delivery Management Module serves as the primary operational component of the proposed system. This module is responsible for recording, updating, organizing, and maintaining delivery records associated with trucking operations. Authorized users may create, edit, review, and update delivery information including delivery ID, truck plate number, assigned driver, client information, delivery route, delivery date, delivery status, revenue, delivery expenses, and Proof of Delivery (POD) documents.

The module automatically computes delivery profitability by calculating the margin between delivery revenue and recorded expenses. By maintaining organized delivery records, the module supports operational monitoring, transaction documentation, and delivery management within the company.

2. Transactions Management Module

The Transactions Management Module is responsible for recording and managing company expenses that are not directly tied to a specific delivery transaction. This module allows authorized users to add, edit, review, and maintain records related to operational expenses such as insurance, vehicle registration, maintenance, utilities, rent, office supplies, equipment purchases, and other business-related expenditures.

By maintaining accurate records of company expenses, the module supports financial monitoring and provides essential information for profitability analysis and management reporting.

3. Search and Record Retrieval Module

The Search and Record Retrieval Module is responsible for improving accessibility and organization of operational records within the system. This module allows users to quickly search, filter, and retrieve information using delivery IDs, truck plate numbers, driver names, client names, and date filters.

The module provides access to historical delivery records, transaction records, and truck-related information, allowing users to efficiently locate operational data. By simplifying record retrieval, the module supports faster monitoring, improved record management, and more efficient access to company information.

4. Reports and Profitability Analysis Module

The Reports and Profitability Analysis Module is responsible for generating operational and financial reports for monitoring, documentation, and management review. This module allows users to create reports based on recorded deliveries, revenue, expenses, and profitability data.

Generated reports may include delivery activity reports, revenue reports, expense reports, profitability reports, and operational summaries. The module also supports printable reports and export functions for documentation and reporting purposes. By consolidating operational and financial information, the module assists management in evaluating business performance and profitability.

5. Security and Administration Module

The Security and Administration Module is responsible for maintaining secure access to the system and managing user accounts and permissions. This module implements login authentication and role-based access control to ensure that only authorized personnel can access specific system functions and records.

The module also records user activities such as record creation, modification, deletion, and login history through activity logs and audit trails. In addition, administrators may manage user accounts, assign roles, and monitor system usage. By controlling access and maintaining activity records, the module helps ensure security, accountability, and protection of operational information within the organization.

Platform

- I. Operating System - Windows
- II. Usage
 - a. Online
- III. Software Specifications:
 - 1. Frontend: Next.js + React + TypeScript + Tailwind CSS + shadcn/ui + Lucide React
 - 2. Backend: Next.js Server Actions + Route Handlers
 - 3. Database: Postgres SQL via Supabase
 - 4. File Storage: Supabase
 - 5. Auth: NextAuth.js (Auth.js)
 - 6. Deployment: Vercel
- IV. Hardware Specifications (minimum specs)
 - 1. Processor: Dual-core 2.0 GHz or higher.
 - 2. RAM: 8 GB
 - 3. Storage: At least 1 GB of available disk space
 - 4. Display: 1366 × 768 screen resolution or higher
 - 5. Browser: Any modern web browser
 - 6. Connectivity: Stable internet connection